

MAANAS TYAGI

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PROFESSIONAL SUMMARY

AI/LLM engineer with hands-on experience building production automation systems and evaluated retrieval-augmented generation (RAG) pipelines. Proficient in RAG, natural language processing (NLP), and agentic AI architectures. Author of one preprint and one paper currently under review on large language model (LLM) behavior.

SELECTED IMPACT

- Demonstrated through a 33-query, RAGAS-scored ablation study that semantic-only retrieval outperforms hybrid BM25+semantic retrieval on every metric (Recall@5: 0.848 vs. 0.818, MRR: 0.703 vs. 0.610), overturning the assumed hybrid-retrieval default
- Replaced a \$3,000 licensed reconciliation tool with a custom Python automation pipeline at Genpact, reducing 10,000-row file processing time from over 30 minutes to under 5 minutes for Fortune 500 clients; now actively used by tens of colleagues to automate their daily reconciliation work
- Contributed 5 merged pull requests to Hugging Face repositories (tau: #314, #290, #289; Transformers: #45150), each reviewed and accepted by project maintainers
- Designed and built end-to-end AI systems at Sstudize, including system architecture and the application layer for Student GPT, along with a token economics model for sustainable per-user LLM cost structure
- Co-built TrenTorch to 130+ GitHub stars in 2 days of launch — one of the fastest-growing open-source educational ML repositories on GitHub, with 10+ forks

PROFESSIONAL EXPERIENCE

LLM Engineer

July 2026 – Present

Sstudize · Remote

- Architect and own all AI systems for the department, including system design and the application layer for Student GPT, an AI-powered platform for students
- Developed a token economics model to define sustainable per-user LLM cost and usage structure for the product

AI & Analytics Intern (*Forward Deployed Engineer Level Work*)

May 2026 – June 2026

Genpact · Jaipur, India

- Replaced a \$3,000 licensed reconciliation tool with a custom Python automation pipeline, reducing processing time for 10,000-row files from over 30 minutes to under 5 minutes; now actively used by tens of colleagues for daily reconciliation work
- Built purchase, invoice, and GRNI reconciliation engines for Fortune 500 clients, converting manual pivot-table workflows into single-file uploads with instant, fully cross-matched output

PUBLICATIONS

- **Instruction Fidelity Decay in Small Language Model Agents: Semantic Recall Dissociates from Format Compliance Under Semantic Interference** (Under Review)
- **Front-Loading Bias in Large Language Models: A Multi-Model Empirical Analysis** (ResearchGate Preprint) – doi.org/10.13140/RG.2.2.32504.92164

PROJECTS

RAG Eval System | LangChain · Chroma · RAGAS · Python

July 2026

github.com/maanas1234/RAG-Eval-System

- Constructed a retrieval and generation evaluation harness using Precision/Recall/MRR and RAGAS faithfulness/relevancy metrics, tested against 33 hand-labeled queries over a 69-document corpus, scoring retrieval and generation quality independently
- Ablation study: semantic-only retrieval outperformed hybrid BM25+semantic retrieval on every metric (Recall@5: 0.848 vs. 0.818, Faithfulness: 0.788 vs. hybrid), overturning the assumed hybrid-is-better default
- Eliminated multi-query retrieval after it matched or underperformed on every metric while approximately doubling LLM calls per query; evaluated k=3, k=5, and k=8, selecting k=5 as the optimal precision/recall/answer-relevancy tradeoff

Personal Intelligence System | Python · FastAPI · OpenRouter · Multi-Agent

Ongoing

- Built a self-hosted personal knowledge engine with 13 plug-and-play connectors (Notion, Slack, Email/IMAP, GitHub, Jira, Linear, Confluence, and more) that chunk, embed, and store all ingested content in a semantic vector space queryable in natural language
- Architected a two-layer agent system: Boss Agent auto-schedules syncs across all connected sources; Brain Agent dispatches to per-source sub-agents with retry/backoff and hash-based content dedup to skip unchanged items

SLM Pretraining From Scratch | PyTorch · Python

June – July 2026

- Implemented a decoder-only transformer (GPT-style architecture) from scratch, including attention mechanism, tokenizer, and training loop, without pretrained weights or external frameworks; 2.1M parameters
- Trained to a validation loss of 1.74 (~5.7 perplexity) using cosine learning rate decay with warmup; evaluated 5 architecture configurations from 120K to 4.8M parameters to characterize the relationship between model size and overfitting in a low-data regime

TrenTorch | Python · NumPy · OSS

Aug 2026 – Present

- Core engineer on TrenTorch (github.com/TrenTorch/TrenTorch) — open-source educational framework teaching PyTorch internals by rebuilding every component from scratch; own the tren CLI layer: test runner, Jupyter integration, progress tracking, benchmarking

EDUCATION

Vellore Institute of Technology

Vellore, India

Bachelor of Technology in Information Technology, 2024 – 2028

SKILLS

Languages: Python, SQL, C

RAG / Retrieval: LangChain, RAGAS, Precision/Recall/MRR, CRAG, Chroma, FAISS, Semantic Search

Modeling: PyTorch, Transformer Architecture (from scratch), Fine-Tuning (LoRA+PEFT), Prompt Engineering

Backend / Tools: FastAPI, Flask, React, Pandas, NumPy, Docker, Git, PostgreSQL, MySQL